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**NMAM INSTITUTE OF TECHNOLOGY, NITTE**  
**Off-Campus Centre of Nitte (Deemed to be University)**  
**Fourth Semester B.Tech (CBCS) Degree Examinations**

May 2025

**CS2102-1 – DATABASE MANAGEMENT SYSTEMS**

(For CS, IS, AM, CC, AD)

Duration: 3 Hours

Max. Marks: 100

**Note:**

**Part – A: Multiple Choice Questions:** Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

**Part – B: Descriptive Answer Questions:** Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

**PART - A: MULTIPLE CHOICE QUESTIONS**

**20 Marks**

1. Which of the following best describes a database?  
A) A collection of unrelated data  
B) A structured collection of related data stored for retrieval  
C) A set of tables without any relationships  
D) A file system
2. In ER-to-Relational mapping, what does each entity type translate into?  
A) A relationship  
B) A foreign key  
C) A table  
D) A schema
3. Which type of attribute cannot be divided into smaller parts?  
A) Composite attribute  
B) Derived attribute  
C) Multivalued attribute  
D) Simple attribute
4. Which of the following attributes is a derived attribute?  
A) Date of birth  
B) Age  
C) Name  
D) Address
5. In an ER model, what does a diamond symbol represent?  
A) Attribute  
B) Entity  
C) Relationship  
D) Primary Key
6. A relationship where an entity is related to itself is called:  
A) Binary Relationship  
B) Recursive Relationship  
C) Ternary Relationship  
D) Self-Relationship
7. For each attribute of a relation, there is a set of permitted values, called the \_\_\_\_\_ of that attribute.  
A) Domain  
B) Relation  
C) Set  
D) Schema
8. What SQL statement would you use to create SQL database named Test?  
A) CREATE DBT Test;  
B) CREATE Test  
C) CREATE DATABASE Test;  
D) MAKE DATABASE Test;
9. How can you change "Thomas" into "Michel" in the "LastName" column in the Users table?  
A) UPDATE User SET LastName = 'Thomas' INTO LastName = 'Michel'  
B) MODIFY Users SET LastName = 'Michel' WHERE LastName = 'Thomas'  
C) MODIFY Users SET LastName = 'Thomas' INTO LastName = 'Michel'  
D) UPDATE Users SET LastName = 'Michel' WHERE LastName = 'Thomas'
10. Which command is used to change the definition of a table in SQL?  
A) CREATE  
B) UPDATE  
C) ALTER  
D) SELECT

11. What is the result of the query **SELECT COUNT (DISTINCT Salary) FROM EMPLOYEE;**?
- A) Total number of employees  
B) Total number of unique salaries  
C) Total number of salaries greater than a specified value  
D) Average salary of employees
12. **EXISTS operator** is used with **WHICH** of the following clause(s)?
- A) Select  
B) In  
C) Where  
D) Group by
13. What is the primary use of a SQL view?
- A) To store permanent data in the database  
B) To create a virtual table from a SQL query  
C) To create relationships between tables  
D) To perform database migrations
14. What happens when **DROP TABLE CASCADE** is executed?
- A) Only the table structure is removed  
B) The table and its dependent objects are removed  
C) The table data is cleared, but the structure is preserved  
D) The table is locked for further operations
15. Which normal form ensures that no non-prime attribute is dependent on a part of a candidate key?
- A) First Normal Form (1NF)  
B) Second Normal Form (2NF)  
C) Third Normal Form (3NF)  
D) Boyce-Codd Normal Form (BCNF)
16. What is the time complexity of searching in a B+ Tree?
- A)  $O(\log n)$   
B)  $O(n)$   
C)  $O(1)$   
D)  $O(n \log n)$
17. Which indexing method is best for dynamic datasets?
- A) Hash Indexing  
B) Bitmap Indexing  
C) B+ Tree Indexing  
D) Sorted File
18. To generate the output of a query, a \_\_\_\_ is responsible
- A) Query execution engine  
B) Query execution motor  
C) Query execution train  
D) None
19. The "all-or-none" property is commonly referred to as \_\_\_\_
- A) Isolation  
B) Durability  
C) Atomicity  
D) None of the mentioned
20. A transaction that has not been completed successfully is called as \_\_\_\_
- A) Compensating transaction  
B) Aborted transaction  
C) Active transaction  
D) Partially committed transaction

### PART - B: DESCRIPTIVE ANSWER QUESTIONS

| Unit – I |                                                                                                                  | Marks | BT* | CO* | PO |
|----------|------------------------------------------------------------------------------------------------------------------|-------|-----|-----|----|
| 1.       | a) What are the Characteristics of Database approach? Explain.                                                   | 8     | L*2 | 1   |    |
|          | b) With a neat diagram, explain the three Schema architecture.                                                   | 8     | L2  | 1   |    |
| 2.       | a) Explain the following types of attributes with examples:<br>(i) Composite<br>(ii) Multivalued<br>(iii) Atomic | 6     | L2  | 1   |    |
|          | b) Explain the Cardinality Ratio for Binary relationships with examples.                                         | 10    | L2  | 1   |    |
| 3.       | a) Consider the following Tables:                                                                                |       |     |     |    |

STUDENT(SName, SUSN, SPhoneNo, SAddress, SAge, SCGPA)  
COURSE (Ccode, Ctitle, SUSN, Sname, Cstaff)

- (i) What is the Degree of the relation STUDENT?  
 (ii) What is the Degree of the relation COURSE?  
 (iii) Construct a Relational algebra to display all details of students with SCGPA>8.0.  
 (iv) Construct a Relational algebra to display names of all students who have taken up 'DBMS' course.  
 (v) Construct a relational algebra to display the names of all courses taken up by students with SCGPA>9.0.
- b) Explain the steps in ER to Relational Mapping Algorithm.

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|----|----|---|---|
| 10 | L3 | 2 | 1 |
| 6  | L2 | 2 | 1 |

**Unit – II**

4. a) Explain how you can Specify the Not Null, DEFAULT and CHECK constraints in SQL.  
 b) Explain the CASCADE and RESTRICT options of DROP TABLE command.  
 c) What are the uses of ALTER TABLE command?
5. a) What are the Informal Design Guidelines for relational schema? Mention all and explain any 2.  
 b) What is a Functional Dependency? Explain with an example.
6. a) Explain the Standard Inference rules for Functional Dependencies.  
 b) Explain the algorithm for determining X+, the Closure of X under F.

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|----|----|---|---|
| 6  | L2 | 2 | 1 |
| 6  | L2 | 2 | 1 |
| 4  | L1 | 2 | 1 |
| 8  | L1 | 2 | 1 |
| 8  | L2 | 2 | 1 |
| 6  | L1 | 2 | 1 |
| 10 | L2 | 2 | 1 |

**Unit – III**

7. a) What is a B+ Tree? Explain structure & characteristics of a B+ Tree with a neat diagram.  
 b) Briefly explain the ACID properties.
8. a) What is a Query Optimizer ? Explain its working with a neat diagram.  
 b) What are the 2 rules of Strict 2PL ? Briefly explain.

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|----|----|---|---|
| 10 | L1 | 3 | 1 |
| 6  | L2 | 3 | 1 |
| 10 | L1 | 3 | 1 |
| 6  | L2 | 3 | 1 |

BT\* Bloom's Taxonomy, L\* Level; CO\* Course Outcome; PO\* Program Outcome.

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